Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/01/2014
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	In vivo induction of chromosome instability and aberrant patterns of DNA methylation in hematopoietic stem/progenitor cells by heavy ions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Kanokporn.Rithidech@sbumed.org	**Fax:**	FY (631) 444-3424
PI Organization Type:	UNIVERSITY	**Phone:**	(631) 444-3446
Organization Name:	State University New York at Stony Brook		
PI Address 1:	Department of Pathology		
PI Address 2:	BHS T9 Health Sciences Center		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	07/01/2011	**End Date:**	06/30/2015
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 6/30/2015, per NSSC information and PI (Ed., 4/22/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Whorton, Elbert Ph.D. (Galveston National Laboratory)		
Grant/Contract No.:	NNX11AK91G		
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Performance Goal Text:			

Task Description:	Cancer of blood cells, in particular myeloid leukemia, is one of the major health concerns from exposure to space radiation. However, the predictions of risks for developing cancer of blood cells after exposure to space radiation are unsatisfactory. To increase the reliability of predicting risk for developing cancer of blood cells, a much improved understanding of space radiation-induced changes in the target cells (stem cells and their progenitors) in living organisms for the induction of this particular type of cancer is critically important. Since human experiments are not possible, animal studies are critically important. In this project, we used the CBA/CaJ mouse as an experimental model to investigate the biological effects of heavy ions. To do this, we exposed mice to two types of heavy ions found in space and relevant to NASA missions, i.e. silicon and titanium. We are conducting experiments to determine the effects of these ions on the induction of damage at the genetic level by measuring the frequency and type of initial- and late-occurring chromosome instability (normally known as genomic instability). Such analyses are being done in hematopoietic stem/progenitor cells (HSPCs) residing in the bone marrow (BM) of exposed mice collected at different times post-irradiation using a novel genome-wide multicolor fluorescence in situ hybridization (whole genome mFISH). The measurement of chromosome instability in HSPCs is coupled with the characterization of damage at the epigenetic level (i.e. the damage affecting gene function without a change in the sequence of DNA molecules) by determining the pattern of DNA-methylation. These two biological endpoints are highly relevant for assessing cancer risks; however, they have not been applied with in vivo exposure to space radiation with silicon or titanium ions. Hence, our results will provide high-priority information for NASA. Our research is the first to use two in vivo endpoints relevant to cancer induction in the HSPCs that are the target of radiation-induced myeloid leukemia (ML) to evaluate the effects of space-radiation. Our data will not only contribute to reducing uncertainties in the assessment of cancer risk from exposure to radiation during space flights but will also improve our understanding of the molecular basis for the influence of space radiation on genomic instability and aberrant DNA-methylation profiles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our data will provide important information that will lead to a better understanding of the patterns of radiation-induced genetic and non-genetic (epigenetic) damage in HSPCs that trigger the induction of genomic instability, a key event of carcinogenesis. This study will couple molecular mechanisms with quantitative data from a robust and highly sensitive technique. The resulting data will deliver information that is of high priority for NASA because the data will not only contribute to reducing the uncertainties in the assessment of cancer risk from exposure to radiation (heavy ions) during space flights but will also improve the understanding of the molecular basis for radiation-induced genomic instability. Hence, the obtained data are critically important for understanding risks associated with space travel as well as developing advanced radiation (heavy ions) therapies for cancer treatments on Earth. Moreover, the information obtained from this project will have a significant impact on the future development of a novel mechanism-based therapeutic strategy to prevent or mitigate the acute or long-term effects of radiation on cells at risk for ML induced by radiation found in space, as well as that found on Earth.
Task Progress:	This project was initiated in October 2011. Below is the list of the work related to this project: 1. Exposure of mice to different doses of 28Si or 48Ti ions. 2. Collection of BM cells from sham controls and exposed mice. 3. Determination of the number and types (according to morphology) of the BM-derived HSPCs colonies. 4. Collection of the BM-derived HSPCs obtained from sham controls and exposed mice for the analyses of chromosome aberrations (CAs) and DNA methylation induced in vivo. 5. Determination of the pattern of DNA methylation (both 5mC and 5hmC) in HSPCs obtained at various times post-irradiation from sham controls and mice exposed to 28Si or 48Ti ions. 6. Measurement of the frequencies of CAs in HSPCs obtained at various times post-irradiation from sham controls and mice exposed to 28Si or 48Ti ions. We have accomplished goals 1 to 4. We have made substantial progress in determining the global (total) levels of 5mC and 5hmC and the frequencies of CAs in HSPCs (goals 5 and 6). The kinetics studies of 5mC and 5hmC after exposure to 28Si or 48Ti ions have been completed and reported last year. Our data suggest that the global level of 5-hydroxymethyl cytosine may be a better predictor of exposure to HZE particles found in space that, in turn, might be more useful for risk assessment than the global level of 5-methyl cytosine. In this annual report, we found reduced expression levels of Ten-Eleven-Translocation 2 (TET2) enzyme (the enzyme known to convert 5mC to 5hmC) in plasma of exposed mice. Such a finding suggests an important role of TET2 in the production of 5hmC. Importantly, our data also suggest that targeting TET2 enzyme (by increasing its expression) may prevent or mitigate the acute or long-term effects of radiation on cells at risk for ML induced by radiation found in space. During the next year of the project, we plan to further measure the expression of other TET enzymes, in particular TET1, and DNA methyltransferases or DNMTs. Ultimately, we will assess if the global levels of 5mC or 5hmC are correlated with chromosomal instability, as a function of 28Si- or 48Ti-ion dose and time post-irradiation. For the cytogenetic analysis (by means of the mFISH method), our current focus is on the delayed- or late-occurring CAs because it is well recognized that it is the best-established end point of radiation-induced genomic instability (a fundamental mechanism known to elevate cancer risk). We have completed the analysis of CAs in HSPCs collected at 6 months after exposure of CBA/Ca mice to silicon or titanium ions. Several types of aberrations were observed in HSPCs collected from exposed CBA/CaJ mice. These include abnormal cells, chromatid breaks, chromosome breaks, and translocations (exchanges). We found dose-dependent increases in the frequencies of abnormal cells, chromatid and chromosome breaks (Analysis of Variance, $p<0.05$, respectively). Of note, our data also showed significant increases in the frequencies of exchanges in HSPCs collected from mice exposed to a medium (0.25 Gy) or a high (0.5 Gy) dose of silicon or titanium ions. Overall, delayed- or late-occurring chromosomal damage was clearly observed in HSPCs collected from mice exposed to silicon or titanium ions, suggesting the occurrence of genomic instability. Since genomic instability is a fundamental mechanism known to elevate cancer risk, our data provide high-priority and highly relevant information for NASA. Although the mFISH method is robust and capable of measuring the frequency, the degree of complexity, and the dose

	response for the production and persistence of simple and complex aberrations over time, it is very time consuming. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide CAs by mFISH is our current major effort and will be continued into next year.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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